- The demand for consistent observability across all operating systems.
- Integration issues that proprietary software/solutions were unable to provide solutions for.

IT teams began utilising open source projects as these gave them the level of flexibility and functional capability needed to solve and support the issues that were challenging proprietary solutions.

The development and deployment of telemetry standards, cross-platform agents, and API-based architectures allowed IT organisations to unify their visibility across multiple systems along with their automated processes.

Lessons learnt from operating open source at scale

There are many benefits of using open source tools but there are also a few lessons learned from implementing these tools in large enterprises.

Complexity when integrating tools: Toolchains can be complex to manage; therefore, it is essential to carefully manage the toolchain components, including data flows, dependencies, and interoperability.

Team readiness: The key to successful implementation of open source tools is not just having the right tools, but also having teams that fully understand how to use them and are able to support, grow, and evolve these tools into something that is effective.

Governance and security: Open source tools provide you with an increased access to information through transparency; however, they also require you to have clear policies and procedures related to patching, version management, and compliance.

Above all, organisations must recognise that flexibility and standardisation should be balanced. If organisations customise their open source tools excessively, they will jeopardise their ability to maintain and sustain these tools, whereas enterprises that implement open source tools in a rigid manner will ultimately lose the ability to adapt to future demands in their business.

The cultural shift behind technical change

Technology alone does not transform operations. The move towards open source-driven observability and automation is closely tied to organisational and cultural change.

Teams must increasingly collaborate across traditional boundaries. Developers, operations engineers, and SREs share the responsibility for reliability. Incident response becomes a learning rather than a blame exercise.

Knowledge sharing extends beyond the organisation. Open source communities provide not just software, but operational patterns, troubleshooting insights, and collective experience.

In this model, maturity is measured not by the number of tools deployed, but by how effectively teams use them to learn and improve.

What IT leaders should evaluate next

For enterprises continuing this journey, several considerations stand out:

Evaluation focus	What it means in practice	Why it matters
Maturity over adoption	Assess whether tools align with defined operational goals rather than adding new technologies for coverage alone	Prevents tool sprawl and ensures observability investments translate into actionable outcomes
Standards over silos	Favour open telemetry standards and interoperable data models instead of isolated, proprietary formats	Reduces long-term integration friction and preserves architectural flexibility
Scale from the start	Design observability and automation with growth, data volume, and complexity in mind	Avoids re-architecture as environments expand across platforms and clouds
Outcomes over trends	Measure success through reliability, performance, and operational resilience — not market hype	Keeps focus on business and service-level objectives rather than tooling popularity

These evaluations help organisations avoid common pitfalls while extracting meaningful value from open ecosystems.

To sum up, open source has transitioned from providing support to serving as a foundational component of the IT workspace. As infrastructure evolves towards greater levels of distribution, automation, and the use of multiple platforms, tools or integrations built using open standards will facilitate observable and automated functionality to ensure reliability at scale. END

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